Mobility and irreducible general isotropy give a Chow-type G3 criterion

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Allowed standard prerequisites (editorial).

- Classical formal geometry G2 finite-cover reduction. Badescu,Projective geometry and formal geometry(2004),Corollary9.20; cited by the author at native672 and retained in the original bibliography. External named G2-to-finite-cover theorem; not an omitted new/current local proof.
- Stein factorization and basic finite/etale family geometry. Classical algebraic geometry; author Lemma2.9 uses Stein factorization. The entire current Lemma2.9 argument is retained. Its base-change/isotropy conclusions are not replaced by this prerequisite.
- Speiser criterion and intersection theory. Original background native314–323, citing Hartshorne Ample Subvarieties(1970); original full bibliography; deformation-invariant intersection products. Named external equivalence used after G3 is established; local Lemma2.6 and the current ramification argument are fully retained.

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Copyright by the author(s). Original article and the adapted article-content portion here are distributed under CC BY-SA 4.0. Changes: selection of Theorem 3.1 with complete author proof, necessary definitions and complete local supporting lemma proofs; independent reformatting and editorial source-location notes. Mathematical wording, formulas and proof steps in the selected author spans are unchanged. No author endorsement is implied; the license supplies no warranties and no additional reuse restrictions are imposed.

One main result is collected. Its two authored cases, the cohomological-dimension conclusion and supporting lemmas are not separate counts. Main statement: native source lines 658–669; proof: 671–703. Core Lemma 2.9: 559–582. The complete current local supporting arguments are retained below, rather than substituted by prerequisite citations.

Source difference retained verbatim. Native line 686 writes $\mathcal{Y}' \subset \mathcal{Y} \times_X \mathcal{Y}$; the author's diagram at line 679 writes $\mathcal{Y}' \subset \mathcal{Y} \times_X X'$, and Lemma 2.9 uses the latter base-change product. Both printed forms remain as supplied; no mathematical correction or correctness certification is made here.

References to the unselected source Proposition 2.10(i) and Lemma 3.8 retain their original source numbers. They appear only as contextual pointers; they are not substituted for the collected core proof. The complete fixed article remains available at https://epiga.episciences.org/19406/pdf. This material has not yet been compiled by this collector.

2 Movable and generating subvarieties

2.1 Background

In this article, X stands for a complex, smooth projective variety. For a connected subvariety Y defined by the ideal $\mathcal{I}_Y \subset \mathcal{O}_X$, let $\hat{X}_Y$ be the formal completion of X along Y; its structure sheaf is $\varprojlim_m \mathcal{O}_X/\mathcal{I}_Y^m$. The rational formal functions $\mathbb{C}(\hat{X}_Y)$ on X along Y form a field. Hironaka–Matsumura [HM68] introduced the following generating properties:

- Y is G2 in X if $\mathbb{C}(X) \hookrightarrow \mathbb{C}(\hat{X}_Y)$ is a finite field extension.
- Y is G3 in X if $\mathbb{C}(X) \hookrightarrow \mathbb{C}(\hat{X}_Y)$ is an isomorphism.

For a detailed treatment of these topics, the reader may consult the reference Bădescu [Bad04]. The following well-known result due to Speiser relates the G2- and G3-properties.

Theorem (*cf.* [Har70])**.** *The following conditions are equivalent* (*for a closed subscheme* $Y \subset X$)*:*

(i) *Y is G3 and Y intersects all divisors of X;*

(ii) *Y is G2 and* $\operatorname{cd}(X \setminus Y) \leqslant \dim X - 2$. (*Here* $\operatorname{cd}(\cdot)$ *stands for "cohomological dimension".*)

The author [Hal20] introduced a sequence of partial amplitude conditions for a subvariety, building upon work of Ottem [Ott12]; the weakest was called $1^{>0}$ (one-positivity).

Definition 2.1 (*cf.* [Hal20, Proposition 1.6(iii)])**.** A locally complete intersection (lci) subvariety Y is $1^{>0}$ if $\mathcal{N}^\vee_{Y/X} = \mathcal{I}_Y/\mathcal{I}_Y^2$ is not pseudo-effective and $\operatorname{cd}(X \setminus Y) \leqslant \dim X - 2$.

Recall, see [Hal19, Corollary 2.6], that the non-pseudo-effectiveness of $\mathcal{N}^\vee_{Y/X}$ implies that Y is *G2* in X. Together with Speiser's criterion, this implies that lci, $1^{>0}$ subvarieties possess the G3-property. The drawback of this definition is that it requires controlling the cohomological dimension of $X \setminus Y$, which is not straightforward. (See [HT17] for various methods to estimate it.)

To motivate the forthcoming mobility considerations, let us also mention that the author proved, see [Hal19, Proposition 3.5], that strongly movable lci subvarieties – this is a simple geometric condition, see below – have non-pseudo-effective co-normal bundle.

2.2 Mobility

For curves and divisors, the concept is classical. In higher codimensions, the notion of (very) moving subvarieties was introduced by Voisin [Voi10], in the attempt to geometrically characterize big subvarieties. A notion with similar flavour, that of generating subvarieties, appears in Chow [Cho69]; see also Debarre [Deb96], Bădescu [Bad04]. The latter notion turns out to be essential for investigating the G3-property of subvarieties of homogeneous varieties. With this motivation, we consider the following.

Definition 2.2.

(a) Let S, the parameter space, be a quasi-projective variety, and let

$$\begin{array}{ccccc} & & \mathcal{Y} & & \\ & \overset{\varpi}{\swarrow} & & \overset{\rho}{\searrow} & \\ S & & & & X \end{array} \qquad \mathcal{Y} \subset S \times X \tag{2.1}$$

be an irreducible, flat S-family of subvarieties of X; we assume that $\mathcal{Y}$ is normal. We consider the following varieties and schemes (they are of interest whenever non-empty):

(i) Let $\Sigma_1 := S$, $\mathcal{Y}_1 = \mathcal{Y}$, $\varpi_1 := \varpi$, and let $\Delta_{\mathcal{Y}} \subset \mathcal{Y} \times \mathcal{Y}$ be the diagonal.

(ii) The incidence variety $\Sigma_2 \subset S \times S$ is the closure of the locus of the following pairs (s, s'): first, one has $Y_s \cap Y_{s'} \neq \emptyset$, Y_s, $Y_{s'}$; second, (s, s') belongs to an irreducible component containing (s, s), (s', s'); that is, $Y_s \cap Y_{s'}$ deforms (non-equidimensionally) to the "identity intersections" $Y_s \cap Y_s$, $Y_{s'} \cap Y_{s'}$. Below is the formal definition.

Let $\mathbb{S} := \{(s, s') \mid Y_s = Y_{s'}\}$; if S parametrizes distinct subvarieties of X, then $\mathbb{S}$ is the diagonal $\Delta_S \subset S \times S$. We define Σ_2 to be the union of those irreducible components

$$R \subset \overline{(\varpi_1, \varpi)(\mathcal{Y}_1 \times_X \mathcal{Y}) \setminus \mathbb{S}} \subset S \times S \underset{\mathrm{proj}_2}{\overset{\mathrm{proj}_1}{\rightrightarrows}} S \quad \text{such that } \mathrm{proj}_1(R) = \mathrm{proj}_2(R) \subset S$$

(we denote the common image by C) and $R \supset \Delta_C$ (the diagonally embedded C). The reason for eliminating $\mathbb{S}$ is to consider only those Y_s, $s \in S$, which possess non-trivial, intersecting deformations. For movable families (see below), this definition simplifies considerably (see Remark 2.3(ii)).

Let $\mathcal{Y}_2$ be defined as in the diagram below:

$$\begin{array}{ccccccc} \mathcal{Y}_2 := \Sigma_2 \times_{S\times S} (\mathcal{Y}_1 \times \mathcal{Y}) & \longrightarrow & \mathcal{Y}_1 \times \mathcal{Y} & \xrightarrow{(\rho_1, \rho)} & X \times X & \xrightarrow{\mathrm{proj}_2} & X. \\ \downarrow & \searrow^{(\varpi_1, \varpi_2)} & \downarrow{\scriptstyle (\varpi_1, \varpi)} & & & & \\ \Sigma_2 & \xrightarrow{\iota} & S \times S & & & & \end{array} \qquad (2.2)$$

For $o \in S$, let $\{o\} \times \Sigma_o := \iota^{-1}(\{o\} \times S)$ and $\mathcal{Y}_o := \mathcal{Y}|_{\Sigma_o}$, and let $\rho_o := \rho|_{\mathcal{Y}_o}$ – the symbol "$|$" stands for restriction – be the evaluation morphism ρ on those Y_s which intersect Y_o non-trivially.

(iii) Suppose $\Sigma_t \subset S^t$, $\mathcal{Y}_t = (\varpi_1, \dots, \varpi_t)^{-1}(\Sigma_t) \subset \mathcal{Y}^t$ are constructed, with morphisms

$$(\varpi_1, \dots, \varpi_t, \rho_1, \dots, \rho_t) \colon \mathcal{Y}_t \longrightarrow (S \times \cdots \times S) \times (X \times \cdots \times X).$$

Here ϖ_i, ρ_i, $i = 1, \dots, t$, are, respectively, the compositions of ϖ, ρ with $\mathcal{Y}_t \to \mathcal{Y}^t \xrightarrow{\mathrm{proj}_i} \mathcal{Y}$. A diagram similar to (2.2) above allows us to define

$$\begin{aligned} \Sigma_{t+1} &:= (\Sigma_t \times S) \times_{\Sigma_{t-1} \times S^2} (\Sigma_{t-1} \times \Sigma_2) \subset S^{t+1}, \\ \mathcal{Y}_{t+1} &:= (\varpi_1, \dots, \varpi_{t+1})^{-1} (\Sigma_{t+1}) \subset \mathcal{Y}^{t+1}. \end{aligned}$$

There is a forgetful morphism of the $(t+1)^{\mathrm{st}}$ component $\Sigma_{t+1} \to \Sigma_t$, and also a "repeating the last component" morphism $\Sigma_t \to \Sigma_{t+1}$. We call Σ_t the *t-incidence variety of S*; roughly, it consists of sequences $(s_1, \dots, s_t)$ whose successive terms belong to Σ_2. A *$\mathcal{Y}$-chain of length t* is a sequence $Y_{s_1}, \dots, Y_{s_t}$, with $(s_1, \dots, s_t) \in \Sigma_t$.

(b) We define the following properties for the family $\mathcal{Y}$:

(i) The family $\mathcal{Y}$ is *movable* if $\rho \colon \mathcal{Y} \to X$ is dominant.

(ii) The family $\mathcal{Y}$ is *strongly movable* if $(\varpi_1, \rho_2) \colon \mathcal{Y}_2 \to S \times X$ is dominant and its smooth locus does intersect $(\varpi_1, \varpi_2)^{-1}(\Delta_S)$ above some open subset S' of S.

In this situation, for $o \in S'$ and a general point $q \in Y_o$, $\mathrm{d}\rho_{o,q} \colon \mathcal{T}_{\mathcal{Y}_o,(o,q)} \to \mathcal{T}_{X,q}$ is surjective and ρ_o is dominant. We say that Y_o is strongly movable.

The family $\mathcal{Y}$ is *strongly movable in codimension one* (codim-one, for short) if there is an open $O \subset X$ such that $O \subset \rho_2(\mathcal{Y}_2|_{\Sigma_2 \cap (S' \times S)})$ and $\mathrm{codim}_X(X \setminus O) \geqslant 2$.

(iii) The family $\mathcal{Y}$ is *generating* if $\big\{(s, x) \in S \times X \mid \exists t \geqslant 1,\ (s, x) \in (\varpi_1, \rho_t)(\mathcal{Y}_t)\big\}$ contains an open subset of $S \times X$. Equivalently, the last component of $\mathcal{Y}$-chains starting with Y_o, where $o \in S$ is general, sweeps out an open subset $O_o \subset X$.

(iv) The family $\mathcal{Y}$ is *chain connecting in codim-one* if the family $\mathcal{Y}$ is generating and the open subset above has the property $\mathrm{codim}_X(X \setminus O_o) \geqslant 2$.

(c) We say that *the smooth subvariety $Y \subset X$ is very moving* (*cf.* Voisin [Voi10, Definition 0.5]) if the following condition holds:

$\forall$ [generic $p \in Y$ and generic $T \subset \mathcal{T}_{X,p}$ of dimension $\dim Y$],
$\exists$ smooth, nearby, embedded deformation Y' of Y, with $p \in Y'$, such that $\mathcal{T}_{Y',p} = T$.

The points p constitute the *very moving locus of* Y. Here we agree that the parameter space S parametrizes sufficiently many deformations Y' of Y such that the tangent spaces $\mathcal{T}_{Y',p}$, for Y' in $S_p := \rho^{-1}(p)$, satisfy the property above. (Note that S does not necessarily consist of all possible embedded deformations of Y in X. In Section 3.2.3, we consider only translates for a group action.) Voisin's condition means that

$$S_p \xrightarrow{\delta_p} \operatorname{Grass}(\dim Y, \mathcal{T}_{X,p}), \quad Y' \longmapsto \mathcal{T}_{Y',p}$$

is dominant about $Y \in S_p$. Therefore, it remains so for $Y' \in S$ in a neighbourhood of Y; all small deformations Y' of Y are still very moving. By abuse of language, we say that *the family* $\mathcal{Y}$ *is very moving* (after possibly shrinking S).

If S_p is smooth at Y, it is enough to require $\mathcal{T}_{S_p,\varpi(Y)} \xrightarrow{\delta_{\varpi(Y),p}} \operatorname{Hom}(\mathcal{T}_{Y,p}, \mathcal{N}_{Y/X,p})$ to be surjective. Since $\mathcal{T}_{S_p,\varpi(Y)} \subset H^0(Y, \mathcal{I}_{Y,p} \otimes \mathcal{N}_{Y/X})$, this map is simply the homomorphism

$$H^0(Y, \mathcal{I}_{Y,p} \otimes \mathcal{N}_{Y/X}) \longrightarrow \operatorname{Hom}\left(\mathcal{T}_{Y,p}, \mathcal{N}_{Y/X,p}\right), \quad \sigma \longmapsto \mathrm{d}\sigma_p.$$

Let us recall, see [Voi10, Remark 2.6], that one may assume, moreover, that $\mathcal{N}_{Y/X}$ is ample. This is still an open condition under deformations.

(d) The *family* $\mathcal{Y}$ *is smooth at* $p \in X$ if $\mathcal{Y}$ is smooth at every $(\varpi(Y), p) \in \rho^{-1}(p)$ and $\mathrm{d}\rho_{(\varpi(Y),p)}$ is surjective:

$$\sum_{\sigma \in \mathcal{T}_{S_p,\varpi(Y)}} \mathrm{d}\sigma_p(\mathcal{T}_{Y,p}) = \mathcal{N}_{Y/X,p}.$$

We say that p belongs to the *smooth locus of* $\mathcal{Y}$. In complex-analytic language, this means that p is a regular value of ρ. We say that ρ is *smooth over codimension one points of* X – for short, ρ *is smooth in codim-one* – if $\rho(\mathcal{Y})$ contains an open subset O such that $\operatorname{codim}_X(X \setminus O) \geqslant 2$ and $\mathcal{Y}$ is smooth at all $p \in O$.

(e) For short, we write A^{gen} for some open (general) subset of a variety or scheme A.

Remark 2.3.

(i) The formulation "deformations of Y" refers *only* to members of some family $\mathcal{Y} \to S$ containing Y. The family is either given or suitably chosen, depending on the context.

The flatness of $\mathcal{Y} \to S$ is required whenever we invoke the deformation invariance of intersection products in $H^*(X)$ with members of the family. Otherwise, the definitions make sense in general.

(ii) We are interested in families $\mathcal{Y}$ which are movable, at the very least. The differential of ρ is surjective (ρ is smooth) on an open locus $\mathcal{Y}^{\mathrm{gen}} \subset \mathcal{Y}$; for any point $p \in \mathcal{Y}^{\mathrm{gen}}$, $\mathcal{Y} \times_X \mathcal{Y}$ is smooth about (p,p). So there is only one irreducible component of $\mathcal{Y}^{\mathrm{gen}} \times_X \mathcal{Y}^{\mathrm{gen}}$ containing the diagonal. Its projection to both S-factors is dominant; let $\mathbb{Y}_2$ be its closure. Thus we get a "natural" component $\overline{(\varpi_1, \varpi_2)(\mathbb{Y}_2) \setminus \mathbb{S}} \subset \overline{(\varpi_1, \varpi_2)(\mathbb{Y}_2) \setminus \Delta_S} \subset \Sigma_2 \subset S^2$. If Y_s, $s \in S$, are pairwise-distinct subvarieties of X, then $\mathbb{S} = \Delta_S$.

(iii) Note that $\Sigma_2 = S \times S$ as soon as the self-intersection satisfies $[Y]^2 \neq 0$; any two members of the family intersect non-trivially. Thus we have $\mathcal{Y}_t = \mathcal{Y}^t$ and $\Sigma_t = S^t$ for $t \geqslant 1$.

The self-intersection is non-trivial in this situation: for $o \in S^{\mathrm{gen}}$, Y_o is lci, $\mathcal{N}_{Y_o/X}$ is ample, and $2 \dim Y_o \geqslant \dim X$. The excess intersection theorem, see [FL82], yields $\deg_{\mathcal{O}_X(1)}[Y]^2 > 0$.

(iv) Suppose X is homogeneous for the action of a group G. Chow, see [Cho69, Deb96, Bad04], introduced the notion of a generating subvariety $Y \subset X$. One verifies that his definition coincides with ours for $S := G$ and $\mathcal{Y} := G \times Y$ (see Lemma 3.8).

Lemma 2.4.

(i) *For the family* $\mathcal{Y}$*, we have the following implications:*

$$\textit{movable} \iff \textit{generating} \iff \textit{strongly movable} \iff \textit{very moving}.$$

(ii) *The family* $\mathcal{Y}$ *generates* X *if and only if* $(\varpi_1, \rho_t)\colon \mathcal{Y}_t \to S \times X$ *is dominant for some* $t \geqslant 2$.

(iii) *Let* $o \in S$ *and* $p_o, q_o \in Y_o \subset \mathcal{Y}$ *be smooth points.*

- *If* $\widehat{\mathrm{d}\rho}_{p_o}\colon \mathcal{T}_{\mathcal{Y},p_o} \to \mathcal{N}_{Y_o/X,p_o}$ *is surjective, then* $\mathcal{Y}$ *is movable.*
- *If* $\widehat{\mathrm{d}\rho}_{p_o} \oplus \widehat{\mathrm{d}\rho}_{q_o}\colon \mathcal{T}_{\mathcal{Y}\times_S\mathcal{Y},(p_o,q_o)} \to \mathcal{N}_{Y_o/X,p_o} \oplus \mathcal{N}_{Y_o/X,q_o}$ *is surjective, then* $\mathcal{Y}$ *is strongly movable. (Note that* $\mathcal{T}_{\mathcal{Y}\times_S\mathcal{Y},(p_o,q_o)} = \{(\xi,\xi') \mid \mathrm{d}\varpi_{p_o}(\xi) - \mathrm{d}\varpi_{q_o}(\xi') = 0.\}$

Here $\widehat{\mathrm{d}\rho}$ *is the composition of* $\mathrm{d}\rho$ *with the projection* $\mathcal{T}_X|_{Y_o} \to \mathcal{N}_{Y_o/X}$.

In general, the mobility conditions are distinct. Let S, Y be positive-dimensional, irreducible projective varieties, and consider $\mathcal{Y} = X = Y \times S$. The morphism ϖ is the natural projection, and ρ is the identity. Then $\mathcal{Y}$ is movable but not generating: any $\mathcal{Y}$-chain has length one. One checks that $\Sigma_2 = \emptyset$, reflecting that the members of $\mathcal{Y}$ have no non-trivial intersections.

Proof. (i) The first three implications are clear. For the last, the surjectivity of $\mathrm{d}\delta_{o,p}$ at general points $o \in S$, $p \in Y_o$ implies that any $v \in \mathcal{T}_{Y_o,p} \setminus \{0\}$ yields a surjective homomorphism $\mathrm{d}\delta_{o,p}(v)\colon \mathcal{T}_{S_p,o} \to \mathcal{N}_{Y_o/X,p}$. Thus, at nearby points $q \in Y_o$, the evaluation at q is surjective:

$$H^0(\mathcal{I}_p \otimes \mathcal{N}_{Y_o/X}) \supset \mathcal{T}_{S_p,o} \ni \sigma \longmapsto \sigma(q) \in \mathcal{N}_{Y_o/X,q}.$$

(ii) Each $(\varpi_1,\rho_t)(\mathcal{Y}_t)$ is constructible, and we have $S\times X = \bigcup_{t\geqslant 1}\overline{(\varpi_1,\rho_t)(\mathcal{Y}_t)}$, by hypothesis. Since $S\times X$ cannot be written as a countable union of proper subvarieties, for some t, we must have $\overline{(\varpi_1,\rho_t)(\mathcal{Y}_t)} = S\times X$. For the converse, there is nothing to prove.

(iii) The first statement is clear. For strong movability, the infinitesimal deformations of (Y_o,p_o) within $\mathcal{Y}_o$ are given by pointed subvarieties (Y_s,p_s), $p_s \in Y_s \cap Y_o$, and correspond to $\xi \in \mathcal{T}_{\mathcal{Y},p_o}$ such that $\widehat{\mathrm{d}\rho}_{p_o}(\xi) = 0$. The surjectivity of $\widehat{\mathrm{d}\rho}_{q_o}$ yields strong movability at the point $((o,o),(p_o,q_o)) \in \mathcal{Y}_2$, which projects to $(o,o) \in \Delta_S$. Briefly, $(\rho,\rho)\colon \mathcal{Y}\times_S\mathcal{Y} \to X\times X$ transversely intersects $Y_o \times X$ at (p_o,q_o). □

Lemma 2.5. *Suppose that* $\mathcal{Y}$ *is such that the self-intersection satisfies* $[Y]^2 \neq 0$. *Then we have*

$$\textit{movable} \iff \textit{strongly movable.}$$

Proof. The morphism $\rho\colon \mathcal{Y} \to X$ is dominant; by Remark 2.3(iii), we have $\mathcal{Y}_2 = \mathcal{Y}^2$, $\Sigma_2 = S^2$. We deduce that (ϖ_1,ρ_2) is dominant. □

Lemma 2.6. *Let* $\mathcal{Y}$ *be a flat family of subvarieties,* $D \subset X$ *an irreducible divisor. The intersection product* $[Y_s]\cdot[D]$ *is non-zero, so* $Y_s \cap D \neq \emptyset$ *for all* $s \in S$ *in the following situations:*

(i) X *is* $\mathcal{Y}$*-chain connected in codim-one.*

(ii) $\mathcal{Y}$ *is generating* X *and* $D \cap O_o \neq \emptyset$. *Here* $o \in S^{\mathrm{gen}}$ *is such that* $Y_o \not\subset D$ *and satisfies Lemma 2.4* (ii)*: for some* $t \geqslant 2$, $\rho_t\big(\mathcal{Y}|_{\Sigma_t\cap(\{o\}\times S^{t-1})}\big)$ *contains the open subset* O_o *of* X.

(iii) $\mathcal{Y}$ *is movable and* $D \cap Y_o \neq \emptyset$ *for* $o \in S^{\mathrm{gen}}$.

Proof. (i) Take $o \in S^{\mathrm{gen}}$ and suppose $[D]\cdot[Y_o] = 0$. The deformation invariance implies that, for all $s \in S$, we have $[D]\cdot[Y_s] = 0$. Then either $D \cap Y_s = \emptyset$ or $Y_s \subset D$; otherwise, the intersection $D \cap Y_s$ is proper, so the intersection product does not vanish.

Let $O_o \subset X$ be the open subset swept out by $\mathcal{Y}$-chains starting with Y_o. We have $\operatorname{codim}_X(X \setminus O_o) \geqslant 2$; thus $D \cap O_o \neq \emptyset$. By chain connectedness, since $D \cap O_o \neq \emptyset$, there are $o = s_0, s_1, \ldots, s_n \in S$ such that $Y_{s_{j-1}} \cap Y_{s_j} \neq \emptyset$ and $D \cap Y_{s_n} \neq \emptyset$.

The previous step implies $Y_{s_n} \subset D$, so $D \cap Y_{s_{n-1}} \neq \emptyset$, *etc.*; inductively, we deduce $Y_o \subset D$. Since $o \in S$ is generic, it follows that $\rho(\mathcal{Y}) \subset D$, contradicting that ρ is dominant.

(ii) We repeat the argument above for a $\mathcal{Y}$-chain connecting the general points of Y_o and D. If the intersection product vanishes, it follows that $Y_o \subset D$, and we have a contradiction.

(iii) The intersection is proper, because $\mathcal{Y}$ is movable and o is general in S. □

Chain connectedness in codimension one is important; we need conditions ensuring it.

Lemma 2.7. *Let* $\mathcal{Y}$ *be a flat, movable family of subvarieties of* X.

(i) *Let* X *be such that its effective divisors are semi-ample (e.g.* $\operatorname{Pic}(X) \cong \mathbb{Z}$, *or* X *is a minimal Mori dream space). Suppose* $\mathcal{Y}$ *is strongly movable. Then any member* Y *intersects the effective divisors of* X, *and* $\mathcal{Y}$ *is chain connecting in codim-one.*

(ii) *If* $\mathcal{Y}$ *is strongly movable in codim-one, then* $\mathcal{Y}$ *is chain connecting in codim-one.*

Proof. (i) Suppose there is a D with $[Y]\cdot[D]=0$. Since mD is globally generated for some $m\geqslant 1$, Y is contained in a fibre F of $X\xrightarrow{\phi}|mD|$; the same holds for the deformations of Y. Note that $\dim|mD|\geqslant 1$. Since Y is movable, after replacing Y by a deformation, we may assume that F is a regular fibre of ϕ. Then $\mathcal{N}_{F/X}|_Y$ (which is trivial, of rank at least one) is a quotient of $\mathcal{N}_{Y/X}$ (which is $(\dim Y-1)$-ample, by [Hal19]), and we have a contradiction, so $[Y]\cdot[D]\neq 0$.

(ii) Let $\mathcal{Y}\to X$, $S'\subset S$, and $O\subset X$ be as in Definition 2.2(b)(ii). Take $x,x'\in O$; there are $o,o'\in S'$ such that $x\in Y_o$, $x'\in Y_{o'}$. The hypothesis implies that there is a $p\in Y_o$ such that $\mathrm{d}\rho_p\colon \mathcal{T}_{\mathcal{Y}_o,p}\to\mathcal{T}_{X,p}$ is surjective, so $\rho(\mathcal{Y}_o)$ is open about the point (o,p). In particular, the $\mathcal{Y}$-chains of length two starting with Y_o cover an open subset $U_o\subset X$. By repeating the argument, we obtain an open subset $U_{o'}$ swept out by the length two chains starting with $Y_{o'}$. Since $U_o\cap U_{o'}\neq\emptyset$, there is a $\mathcal{Y}$-chain of length at most four connecting x to x'. □

2.3 Irreducible pointed deformation spaces and fundamental groups

Definition 2.8. Let $\mathcal{Y}\xrightarrow{(\varpi,\rho)}S\times X$ be a normal, movable family of irreducible subvarieties. We say that $\mathcal{Y}$ has *irreducible general isotropy* if $\rho^{-1}(x)$ is irreducible for all points x in a non-empty open subset $X^{\mathrm{iris}}_{\mathcal{Y}}\subset\rho(\mathcal{Y})$. Equivalently, the geometric generic fibre of ρ is connected ($\mathcal{Y}$ is normal). For $x\in X$, $S_x:=\rho^{-1}(x)=\{s\mid x\in Y_s\}$ can be viewed as a subset of S. The term "isotropy" is chosen in analogy with the case of group actions (see Proposition 2.10i).

Lemma 2.9.

(i) *The family $\mathcal{Y}$ has irreducible general isotropy if and only if $\mathcal{Y}\times_X\mathcal{Y}$ possesses exactly one irreducible component dominating X (which contains the diagonal $\Delta_{\mathcal{Y}}$).*

(ii) *Suppose $\mathcal{Y}$ has irreducible general isotropy, and let $X'\xrightarrow{f}X$ be a finite (surjective) morphism, with X' irreducible. Then the following properties hold true:*

- *$\mathcal{Y}\times_X X'$ contains a unique irreducible component $\mathcal{Y}'$ which dominates X'. It contains $f^{-1}(Y_s)$ for any $s\in S$ such that $Y_s\cap X^{\mathrm{iris}}_{\mathcal{Y}}\neq\emptyset$.*
- *We have $\mathcal{Y}\times_X U'\subset\mathcal{Y}'$, where $U'\subset X'$ is the étale locus of f.*

Proof. (i) For the necessity, note that the general fibre of $\mathcal{Y}\times_X\mathcal{Y}\to X$ is irreducible. For the sufficiency, consider the Stein factorization $\mathcal{Y}\to\hat{\mathcal{Y}}\to X$ (induced by some completion of $\mathcal{Y}$). Should the degree of the second map be greater than one, $\hat{\mathcal{Y}}\times_X\hat{\mathcal{Y}}$ possesses other components besides the diagonal, determined by pairs of distinct points in the fibres of $\hat{\mathcal{Y}}\to X$. Their pre-image in $\mathcal{Y}\times_X\mathcal{Y}$ would still dominate X.

(ii) The general fibres of $\mathcal{Y}\times_X X'\to X'$ and $\mathcal{Y}\to X$ are isomorphic, which implies that

$$\mathcal{Y}'=\overline{\mathcal{Y}'^{\,\mathrm{open}}}^{\mathcal{Y}\times_X X'},\quad \text{where } \mathcal{Y}'^{\,\mathrm{open}}:=\rho^{-1}(X^{\mathrm{iris}}_{\mathcal{Y}})\times_{X^{\mathrm{iris}}_{\mathcal{Y}}}f^{-1}(X^{\mathrm{iris}}_{\mathcal{Y}}).$$

Furthermore, since $Y_s\cap X^{\mathrm{iris}}_{\mathcal{Y}}\neq\emptyset$ and f is finite, every component of $f^{-1}(Y_s)$ dominates Y_s, hence intersects $f^{-1}(X^{\mathrm{iris}}_{\mathcal{Y}})$.

Now proceed to the second claim. (The argument was provided by the anonymous referee.) The étale locus $U'\subset X'$ of f is Zariski open, irreducible. Then $\mathcal{Y}\times_X U'\to\mathcal{Y}$ is étale, so it is an open morphism. Thus any component of $\mathcal{Y}\times_X U'$ dominates $\mathcal{Y}$ and hence U'. Therefore, $\mathcal{Y}\times_X U'$ is irreducible and is contained in $\mathcal{Y}'$. □

3 A Chow-type G3-criterion and applications

3.1 The criterion

We prove that mobility conditions imply the G3-property. This result is inspired from [Cho69] and [Bad04, Theorem 13.4], which consider subvarieties of homogeneous varieties.

Theorem 3.1. *Let Y_o be a G2-subvariety of X which has the following properties:*

- *It belongs to a flat, movable family $\mathcal{Y}=(Y_s)_{s\in S}$ with irreducible general isotropy.*
- *Any effective divisor of X intersects Y_s, $s\in S^{\mathrm{gen}}$. (Equivalently, by Lemma 2.6(iii), the intersection product of Y_o with any effective divisor of X is numerically non-trivial.)*

Then Y_o is G3 and $\mathrm{cd}(X\setminus Y_o)\leqslant\dim X-2$ in either one of the following situations:

(i) *X is algebraically simply connected* ($\pi_1^{\text{ét}}(X) = \{1\}$).

(ii) $[Y_o]^2 \neq 0 \in A^*(X)/\sim_{\text{num}}$.

Proof. Since Y_o is G2, there exist a normal, irreducible, projective variety X', a G3-subvariety $Y'_o \subset X'$, and a finite, surjective morphism $f\colon (X', Y'_o) \to (X, Y_o)$ such that f is étale in a neighbourhood of Y'_o (*cf.* [Bad04, Corollary 9.20]). Let $D' \subset X'$ be the ramification divisor of f and $D := f_*(D')$; we want $D = \emptyset$, that is, f is étale. Let us consider the diagram

$$\begin{array}{ccccccc}
\mathcal{Y}'_{\overline{\omega}} & \longrightarrow & \mathcal{Y}'_{\omega} & \longrightarrow & \mathcal{Y}' \subset \mathcal{Y} \times_X X' & \xrightarrow{\rho'} & X' \\
\downarrow{\scriptstyle \varphi_{\overline{\omega}}} & & \downarrow{\scriptstyle \varphi_{\omega}} & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle f} \\
\mathcal{Y}_{\overline{\omega}} & \longrightarrow & \mathcal{Y}_{\omega} & \longrightarrow & \mathcal{Y} & \xrightarrow{\rho} & X. \\
\downarrow & & \downarrow & & \downarrow{\scriptstyle \varpi} & & \\
\overline{\omega} := \operatorname{Spec}(\overline{\mathbb{C}(S)}) & \longrightarrow & \omega := \operatorname{Spec}(\mathbb{C}(S)) & \longrightarrow & S & &
\end{array}$$

Here $\mathcal{Y}' \subset \mathcal{Y} \times_X \mathcal{Y}$ is the irreducible component introduced in Lemma 2.9, dominating X'. The ramification divisor of $\varphi\colon \mathcal{Y}' \to \mathcal{Y}$ is $D_1 := (\rho')^{-1}(D') \cap \mathcal{Y}'$. We are going to show that $D_1 \cap \mathcal{Y}'_\omega = \emptyset$; it suffices to prove that $D_1 \cap \mathcal{Y}'_{\overline{\omega}} = \emptyset$, that is, $\varphi_{\overline{\omega}}$ is étale.

Since f is étale about Y'_o, Lemma 2.9 implies that $Y'_o = Y_o \times_X Y'_o \subset \mathcal{Y}'$. Thus Y'_o is a component of $\mathcal{Y}'_o = (\varpi\varphi)^{-1}(o)$ and φ is étale in its neighbourhood. But Y'_o is the specialization of a component Z of the fibre $\mathcal{Y}'_{\overline{\omega}} \to \overline{\omega}$, so $\varphi_{\overline{\omega}}\colon Z \to \mathcal{Y}_{\overline{\omega}}$ is étale (because φ is so about Y'_o). The components of $\mathcal{Y}'_{\overline{\omega}}$ are conjugate under the Galois group $\operatorname{Gal}\big(\overline{\mathbb{C}(S)}/\mathbb{C}(S)\big)$, since $\mathcal{Y}'_\omega$ is irreducible. Thus $\varphi_{\overline{\omega}}$ is étale everywhere and so is φ_ω. This proves that $D_1 \cap \mathcal{Y}'_\omega = \emptyset$.

Lemma 2.9 says that $\mathcal{Y}'$ contains $f^{-1}(Y_s)$ for $s \in S^{\text{gen}}$. We deduce that $D_1 \cap f^{-1}(Y_s) = \emptyset$. Hence $D \cap Y_s = \emptyset$; the subvariety Y_s does not intersect the branch divisor. This contradicts that $Y_s \cap D \neq \emptyset$ for $s \in S^{\text{gen}}$ (or that $[Y_o] \cdot [D] \neq 0$) unless $D = \emptyset$, so f is étale.

(i) If X is algebraically simply connected, then f is an isomorphism, so Y_o is G3 in X.

(ii) Now suppose $d := \deg(f) \geqslant 2$, and let $Y'_o, Y'_1, \ldots, Y'_{d-1}$ be the pre-images of Y_o by f; they are deformation equivalent. Then $[Y_o]^2$ can be computed on the covering space X':

$$d \cdot [Y_o]^2 = [Y'_o] \cdot [f^{-1}(Y_o)] = [Y'_o] \cdot ([Y'_o] + [Y'_1] + \cdots + [Y'_{d-1}]) = d \cdot [Y'_o] \cdot [Y'_1] = 0.$$

This contradicts the hypothesis, so $d = 1$ and f is étale. In both situations, Y_o is G3 and intersects all effective divisors, so Speiser's criterion implies that $\operatorname{cd}(X \setminus Y_o) \leqslant \dim X - 2$. □

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